Machine Learning Based Characterization of a Waterproofed Soft Multi-directional Force Sensor

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***Abstract*—A cost-effective soft multi-directional force sensor can enable next-generation robots to recognize object features by touch, similar to humans. This paper presents a novel approach for force region classification on a soft hemispherical sensor made up of four flex sensors placed 90° apart. By considering a quarter of the hemisphere due to its symmetry, we effectively divided the sensor into three distinct regions where force can be applied. We used the K-means clustering algorithm to classify the region of the applied force, which successfully categorized all 60 test trials into the correct force regions. The effective range for detection was about 1.5 - 5 N of force. Our experimental results also showed that the sensor responded well to being kept submerged in salt water for 7 days, indicating its suitability for operation in underwater environments. This study opens up possibilities for 3D multi-directional force recognition using the soft sensor, which has potential usage in diverse fields, such as robotics and healthcare, even in challenging underwater environments.**

***Index Terms*—soft tactile sensing, multi-directional force sensor, cost-effective, waterproof, machine learning**

I. Introduction

In order to interact with the surroundings effectively, robots should be capable of perceiving numerous features of a miscellaneous item promptly just by touching it [1]. However, most of the robotic end-effectors developed so far are predominantly rigid in nature, which lacks the required softness and conformability to grab an object like a human hand. The introduction of soft robotic grippers allows manipulators to overcome this limitation [2], [3]. Nonetheless, sensors to be integrated with these systems need to be soft, flexible, and intelligent enough to receive and process the information readily [4]. This would enhance their grip and control of an unknown object by adjusting their grasp and manipulation techniques in response to feedback acquired by sensors. Targeting this, researchers around the world have been working on different types of soft tactile sensors and methodologies (especially machine learning (ML) techniques) that could provide a sense of touch to the robots by analyzing data collected by the sensors.

There are a decent amount of studies on ML techniques applied for processing data to detect force distribution on a tactile sensor. For example, Sferrazza *et al.* utilized a finite element model to set ground truth data to characterize the three-dimensional force distribution of a vision-based tactile sensor [5]. Yang *et al.* demonstrated a soft tactile sensor that can estimate the contact force, contact feature, and contact point at the same time using ML [6]. They mimicked the pyramid shape of the mountain ridge-like structure in human skin in the sensor to enhance its sensitivity. Al-Handarish *et al.* presented a piezo-resistivity based flexible tactile sensor where carbon black (CB) is dispersed in a microstructure porous sugar/PDMS-based sponge [7]. They employed long short-term memory (LSTM)-dense classification model and five traditional ML methods, such as support vector machine, multilayer perceptron, decision tree, and k-nearest neighbor, to classify different hand gestures integrated with the sensor. Syu *et al.* also adopted the LSTM technique to find out the hand gesture while combining the hand with their fabricated sensor [8]. *Baishya et al.* demonstrated a deep learning method to identify different materials based on the spatio-temporal signal of a flexible tactile skin [9]. Massari *et al.* proposed a Fiber Bragg Gratings (FBGs) based soft tactile sensor which can concurrently determine the magnitude and the location of a normal force applied onto its surface utilizing ML algorithm [10]. Some other examples can also be provided where researchers either developed their own sensor or utilized ML techniques to estimate the contact force magnitude and location [11]–[15]. However, in most instances, the sensor structures are flat, not soft, can measure one-directional force only, and none of them are waterproof. To the best of our knowledge, there is hardly any example of a waterproof soft tactile sensor that can measure multi-directional force in 3D space, especially with the application of ML technique.

In this study, we propose an ML algorithm to classify the region of applied force on a waterproof soft hemispherical force sensor, which would later help determine the force magnitude as well. The sensor consists of four flex sensors embedded in a silicone hemisphere and positioned 90° apart in a 3D-printed O-ring (Fig. 1c(i)). To classify the region, we utilize K-means clustering, an unsupervised ML algorithm for clustering data points into groups or clusters based on similarity. This approach would certainly streamline autonomous and reliable identification of the magnitude and direction of the force on the sensor, ultimately improving the performance of the sensor in various applications.

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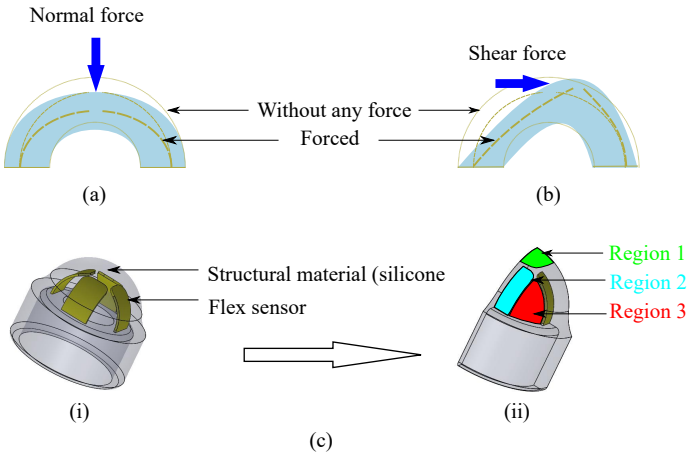


Fig. 1. Working principle of the sensor against (a) normal force and (b) shear force with blue arrows depict applied force directions, and (c) sensor structure: (i) 3D model of the full sensor and (ii) quarter of the hemisphere and designated regions.

II. Methodology

A. Sensor Structure and Working Principle

Our proposed sensor is made of soft silicone (Dragon Skin Fx Pro (DSFP), Reynolds Advanced Materials, Boston, MA, USA) as a structural material. The base of the sensor is made of a 3D-printed O-ring to make the sensor robust against water. Four flex sensors are embedded inside the molded silicone and positioned 90° apart from one another as depicted in Fig. 1(c). If a normal force is applied on the top of the sensor, all four flex sensors would bend similarly and provide a unidirectional output change, whereas different distinctive output patterns for other forces. Tentative deformation of the sensor structure and the flex sensors are exhibited in Fig.1 (a) and (b). Readers are directed to our previous works [16], [17] for a better understating of the sensor structure, working principle, and the fabrication process of the sensor. Here, we consider the quarter of the hemisphere for demonstrating our proposed method since it can easily be replicated for the entire hemisphere. We designate 3 regions in the segment: at the top of the sensor - region 1, on the side of the sensor where a flex sensor is located - region 2, and the space in between the two flex sensors - region 3. In the following sections, we have shown how to distinguish the location of the force among these three regions with the help of the ML technique. For the simplicity of the model, we consider applying only normal forces for the detection of the applied force region since the effectiveness of the sensor for capturing different directional forces has already been discussed in our previous studies [16], [17].

B. Sensor Dataset

As seen in Fig. 1c(ii), three different regions on the sensor were identified as region 1, region 2, and region 3, where forces can be applied. These are the three different classes that set up the classification problem. The sensor dataset is a time series where the four flex sensor values are recorded as the force increases from 0 to 5 N. The dataset has five columns for each trial: one for time and four for the flex sensor values. The dataset had 30 trials for each of the three classes, making 90 total trials. It was split into two datasets: i) calibration dataset with 10 trials and ii) testing dataset with 20 trials.

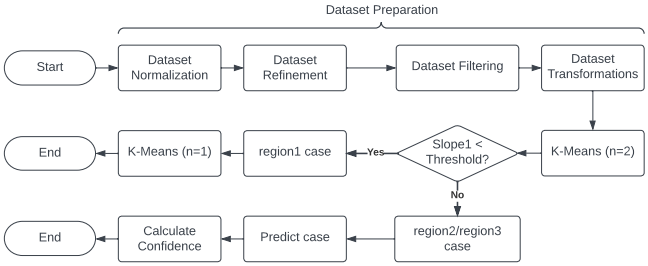


Fig. 2. Flowchart for the force position detector algorithm.

C. Machine Learning (ML) Algorithm

The algorithm determines the region on the sensor where force is being applied. Fig. 2 shows the algorithm flowchart. The first step is dataset preparation. In this step, the algorithm normalizes the sensor values in each trial using the maximum sensor values. After normalization, the initial sensor readings equivalent to 0.5 N force are discarded in the dataset refinement step. Savitzky-Golay filter is used in the next step to smooth the sensor data time series [18]. Finally, the filtered dataset is transformed for the following steps. After dataset preparation, the algorithm has two main phases: i) the calibration phase and ii) the testing phase. The prepared dataset is split into two for these phases. The algorithm relies on K-means clustering in both phases. K indicates the number of clusters into which the data should be divided. For region 2 and region 3, the number of sensors in each cluster using K=2 easily differentiates between the two regions (Figs. 3(c) and (d)). However, using K-means to separate region 1 from the others is challenging because the smallest meaningful value of K is 2, and considering a small number of data points without knowing the corresponding force will likely result in region 1 being categorized as region 2 or region 3, i.e., having two clusters (as seen in Fig 3(a)). This is why a threshold is used to distinguish region 1 from region 2 and region 3. The threshold is based on the observation that when the sensor value series is divided into two clusters (K=2), region 1's first centroid series would always have a lower slope than both region 2 and region 3, as shown in Figs. 3(a), (c), and (d). This threshold is established using the calibration dataset during the calibration phase. In the testing phase, K=2 is used for the whole test dataset, and region 1 gets isolated using the threshold, as depicted in Fig 2 as well as Fig 3(a) and (b). For region 2 and region 3, as predictions are made at each slice of the time series, a term called confidence is introduced. The final prediction corresponds to the case that is predicted the most. The confidence is calculated based on the number of data points within the time series that have the same prediction as the final projected case. For example, in

Fig 3(c), the prediction is region 2 with a confidence value of 100 %, whereas, in Fig 3(d), the prediction is region 3 but the confidence is less than 100 % since some of the time series data were categorized as region 2 (seen as sharp changes around X_data = 200). X_data in Fig 3 refer to the indices of each sampled sensor datapoints.

III. Results and Discussions

A. Machine Learning (ML) Based Characterization

From the sensor dataset, 30 trials from the three cases (10 each) were used to determine the slope1 threshold. For testing, 60 trials from each case (20 each) were run through the algorithm for prediction. All 60 trials for all three cases were successfully categorized into the correct region. The average confidence of prediction for region 2 was 93.6 %, and 77.3 % for region 3. From the average confidence values, we can get an insight into the minimum forces for recognition of the force region. The average minimum force for region classification is 0.8 N for region 2, and 1.5 N for region 3 for the manufactured sensor. Indeed, we see the initial data points for region 3 in Fig 3(d) are hard to interpret, and the prediction is unstable. This, in turn, decreases the effective average range of the sensor. The range can be increased by minimizing manufacturing deviations such as placing the flex sensors exactly at 90° apart, minimizing the chance of slippage between the flex sensor and the sensor structure, etc.

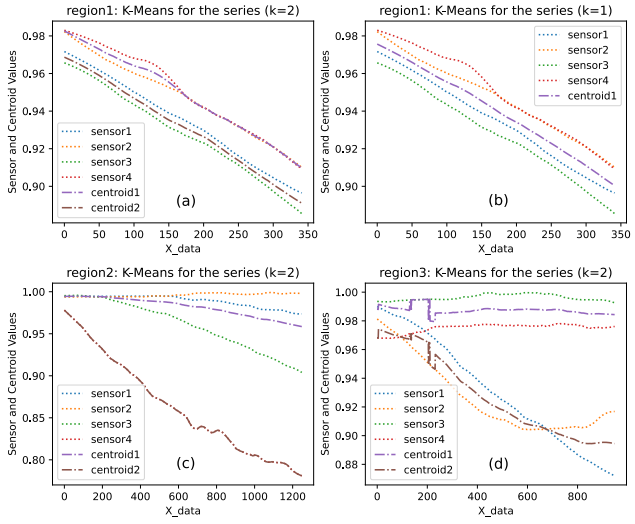


Fig. 3. K-Means with (a) k=2 for region 1, (b) k=1 (threshold condition passed) for region 1, (c) k=2 for region 2, and (d) k=2 for region 3.

B. Waterproofing Tests

Previously, we demonstrated the waterproofing characteristics (IP68 rated) of the sensor for normal water [17]. Here, we report the sensor is good enough to withstand salt water which implies its potential applications in marine environments as well. In order to perform the test, we kept our sensor sample submerged in salt water. The saltwater is taken as a replica of seawater, where the density of the salt is around 35 grams/kg. In order to get the force response, a normal force was applied on the top of the sensor. As shown in Fig. 4, the sensor is working perfectly even after being under salt water for consecutive 7 days.

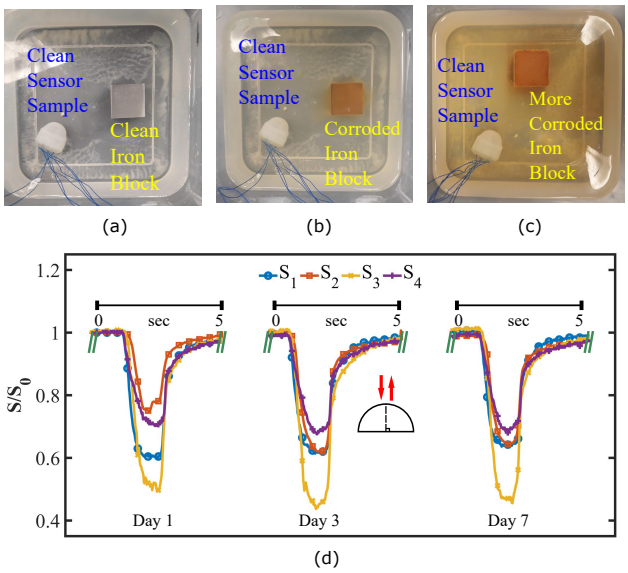


Fig. 4. Waterproofing and corrosion resistance test with a sensor and an iron block in salt water: (a) both samples are clean on Day 1, (b) clean sensor, visibly corroded iron block on day 3, (c) clean sensor, more corrosion on the iron block on day 7, and (d) normal force response of the submerged sensor.

IV. Conclusion

In this study, we present a unique method for classifying applied force regions on a soft hemispherical sensor made up of four flex sensors. The K-means clustering method and a slope threshold were used for classification. The threshold is determined from the calibration dataset and should be calibrated accordingly when a different sensor is used. Our results demonstrated that the algorithm could detect all 60 test trials successfully with an adequate range for detection of about 1.5 - 5 N of force. Based on the results, we can conclude that our soft hemispherical force sensor, combined with the K-means clustering algorithm, is a promising approach for force sensing in various applications. Additionally, the sensor demonstrated reliable performance when tested in saline solution for 7 days, making it applicable to the marine environment.

Our study lays the foundation for further development of advanced machine learning techniques to enhance the sensor's performance. In the future, we aim to integrate the sensor system with suitable robotic grippers for various object recognition. Overall, our soft hemispherical force sensor has potential where accurate and reliable 3D multi-directional force measurements are crucial.

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